



Ryan, Paul McKenna

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Address: 23 Ribble Drive, Biddulph, Staffordshire, ST8 7LL, Stoke-On-Trent, United Kingdom (Home)

ABOUT ME

I am an Embedded Systems Engineer and PhD researcher with a unique blend of deep theoretical expertise and practical, hands-on engineering experience. My professional focus lies at the intersection of **Embedded Linux, kernel driver development, and RTOS-based systems**, with a strong foundation in physics, robotics, and applied machine learning.

I thrive in multidisciplinary environments where software and hardware meet. Whether it's implementing **Linux device drivers**, soldering prototypes, machining parts, or debugging with an oscilloscope, I enjoy working across the stack to deliver robust, integrated systems.

Currently completing a PhD in Computer Science at the University of York—with a research focus in robotics, reinforcement learning, and deformable object manipulation—I bring a high level of mathematical and analytical rigor to engineering challenges. My passion lies in building real-world systems: from **Yocto builds, device trees, and low-level firmware** to schematics, PCB design, and physical prototyping.

WORK EXPERIENCE

 **LINUX - PRINTING TECHNOLOGIES** – CAMBRIDGE, UNITED KINGDOM

Address 8 Stocks Bridge Way, St. Ives, PE27 5JL, Cambridge, United Kingdom

EMBEDDED SYSTEM SOFTWARE DEVELOPER – 02/12/2024 – CURRENT

- Designed, implemented, and maintained **custom Linux kernel device drivers** for printer subsystems, interfacing over **SPI, I²C, and UART** with exposure to **PCIe** and other high-speed interfaces.
- Implemented custom **platform** and **non-platform drivers**, authored **device trees**, and optimized **virtual memory** mappings for embedded Linux subsystem
- Built and optimized **Yocto-based Embedded Linux** distributions; configured **virtual memory**, MMIO, and board bring-up to meet real-time performance and reliability targets.
- Delivered production **firmware in Embedded C/C++** across **Embedded Linux and RTOS-based** subsystems where required, ensuring deterministic timing and safe concurrency.
- Performed **electrical fault-finding and hardware validation** with **oscilloscopes, function generators, thermal cameras**, and logic-level probing; closed issues end-to-end from root cause to fix.
- **Designed and fabricated custom test rigs** (e.g., valves, pumps) including **schematics, PCB prototyping/assembly**, embedded firmware, and verification to accelerate debug and release.
- **Ensured compliance** of embedded solutions with **RoHS, CE, and IEC 61508** (functional safety) requirements.
- Owned the customer interface: scoped defects/enhancements, proposed **low-level solutions**, agreed effort and cost, and delivered to spec and schedule for **industrial automation/printing** used in global logistics and regulatory marking.
- **Developed and tested FPGA logic in VHDL** for custom industrial printer subsystems, integrating hardware accelerators with **Embedded Linux drivers** to enhance real-time performance and reliability.

 **MBDA** – BRISTOL, UNITED KINGDOM

Address 3 Golf Course Ln, Bristol , BS34 7QS, Bristol, United Kingdom

EMBEDDED SYSTEMS SOFTWARE DEVELOPER – 17/02/2022 – 19/04/2023

- Developed **Embedded Linux and RTOS** firmware in **Misra C/C++** for **ground-based air defence** platforms and **radar** subsystems, focusing on high-reliability operation.
- Implemented and hardened **CAN-bus** communication (plus **SPI/I²C/UART**) for distributed embedded nodes; optimized bus utilization, error handling, and real-time message scheduling.
- Created and adapted **kernel modules, device trees, and platform drivers** for custom defence hardware; integrated board support into **Yocto** builds for secure, repeatable deployments.
- Contributed to **embedded networking** (e.g., Ethernet/serial links) for command-and-control data paths; improved interoperability and latency across system components.
- Performed hardware/software bring-up and validation using multi-meter, oscilloscopes, function generators, resistive loads; collaborated tightly with electronics and systems teams under **Defence** constraints.
- Operated under **SC (Security Check) clearance**, adhering to defence engineering processes, configuration control, and safety/certification expectations.

ELIXIR SOFTWARE – MANCHESTER, UNITED KINGDOM

Address Glasshouse, Alderley Park, Congleton Road Nether Alderley, Macclesfield, SK10 4ZE, Manchester, United Kingdom

SOFTWARE ENGINEER – 01/07/2019 – 28/03/2021

- Built high-performance **scientific computing and data analysis tools** in C++/Python, boosting laboratory efficiency and accelerating drug discovery workflows.
- Applied **software engineering best practices** (clean reusable code, statistical modelling, visualization) to deliver reliable, maintainable systems for research applications.

● EDUCATION AND TRAINING

10/10/2023 – CURRENT York, United Kingdom

DOCTOR OF PHILOSOPHY IN COMPUTER SCIENCE University of York

Address Heslington, York, YO10 5DD, York, United Kingdom | **Website** <https://www.york.ac.uk/> | **Level in EQF** EQF level 8

12/09/2019 – 26/06/2022 Chester, United Kingdom

MASTER OF SCIENCE IN MATHEMATICS University of Chester

Address Parkgate Rd, Chester, CH1 4BJ, Chester, United Kingdom | **Website** <https://www.chester.ac.uk/> |

Level in EQF EQF level 8

10/09/2016 – 30/06/2019 Sheffield, United Kingdom

BACHELOR'S OF SOFTWARE ENGINEERING Sheffield Hallam University

Address Howard St, Sheffield City Centre, Sheffield, S1 1WB, Sheffield, United Kingdom | **Website** <https://www.shu.ac.uk/> |

Level in EQF EQF level 7

● LANGUAGE SKILLS

Mother tongue(s): **ENGLISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
SPANISH	B1	B1	A2	A2	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

C++ | CUDA | Embedded C / Embedded Linux / RTOS / Testing Fundamentals / Confident using Raspberry pi | programmable logic controller | Numpy | Matplotlib | PyTorch | Python | AutoCAD and Inventor CAD program | Altium Designer, Xilinx, MS Office, AutoCAD, Micro-vision Keil, | OpenCV | communication Protocols(UART,SPI,I2C,CAN) | Lathe Machining | Gcc, Clang, CMake, Make basics | boring machine ,milling machine tool grinding machine training. | Modern C++ Boost Library | CAD (FreeCAD) | ARM CORTEX | MATLAB | Algorithms & Data-structures | Micro-controllers | STM32 | sympy | Linux | AWS | GCP | Pandas | scipy |

ROS | Isaac-gym | Git | Deep Learning | Machine Learning | Scikit-Learn | Artificial Intelligence | SQL | Data Science | Seaborn | Keras | Computer Vision | Data Analysis | Test-Driven-development | software metrics | UML-diagramming | ER-diagramming | RTOS | NVIDIA Omniverse | PCB-Designing